

BYOD: BUILD YOUR OWN DLM - EFFICIENT AUTOREGRESSIVE-TO-DIFFUSION LANGUAGE MODEL CONVERSION

Anonymous authors

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ABSTRACT

Diffusion language models (DLMs) permit parallel token updates and flexible generation orders, but leading systems rely on costly pretraining or continued full-model training. We ask whether diffusion generation can instead be added to existing instruction-tuned autoregressive models using a single parameter-efficient conversion recipe. Build Your Own DLM (BYOD) replaces causal attention with bidirectional attention and trains only query/value LoRA adapters under a masked denoising objective, leaving 94.2–95.9% of parameters frozen. Without architecture-specific tuning, the same recipe converts Gemma 2 9B, Llama 3.1 8B, Qwen2.5 7B, and Ministral 8B, with each 25,000-update conversion processing at most 102.4M token positions on a single GPU. All four converted models learn to generate coherent responses from fully masked sequences. On a preliminary nine-task evaluation, the strongest converted model reaches 52.2% average accuracy, compared with 51.9% for LLaDA 8B Instruct in the same evaluation harness; sample counts remain mixed for the still-incomplete backbone sweep. Continued training does not improve aggregate downstream accuracy, but improves generation quality when the number of denoising network function evaluations (NFE) is smaller than the output length. Together, these results provide a practical route for researchers to build diffusion language models from existing autoregressive checkpoints using only parameter-efficient adaptation and single-GPU compute.

1 INTRODUCTION

Autoregressive (AR) language models generate a sequence from left to right. This method has proven extremely effective due to its simple learning target and effective scaling capabilities. However, it also has trade-offs: it fixes the generation order, limits the model to unidirectional attention, and predicts only one new token at a time. Diffusion language models (DLMs) instead begin with a corrupted sequence and can iteratively reconstruct multiple positions at a time using bidirectional context (Austin et al., 2021a; Sahoo et al., 2024; Shi et al., 2024). Their number of function evaluations (NFE) can be chosen independently of output length, and early low-probability predictions can be revised. However, training or adapting models for diffusive generation has so far been computationally intensive and therefore expensive.

Our aim is to provide a practical alternative to training a separate diffusion foundation model for every model architecture family, reusing knowledge embedded in existing AR models reduces the data, compute, and financial barriers to experimentation. This emphasis follows calls to treat computational efficiency and access as research outcomes alongside accuracy (Strubell et al., 2019). We also estimate the potential sustainability impact, while recognizing that compute volume alone is not a carbon measurement: emissions depend strongly on hardware, datacenter efficiency, location, and energy supply (Patterson et al., 2021). We report bounded incremental training scale and an explicitly assumption-based operational-emissions scenario, while avoiding claims about measured emissions.

We study this question with BYOD, a generalizable conversion and evaluation pipeline for four instruction-tuned decoder families: Gemma 2 9B, Llama 3.1 8B, Qwen2.5 7B, and Ministral 8B. We selected the latest text-only instruction checkpoint from each family with a comparable parameter count. Multimodal variants are not used to avoid modality-specific training as an additional

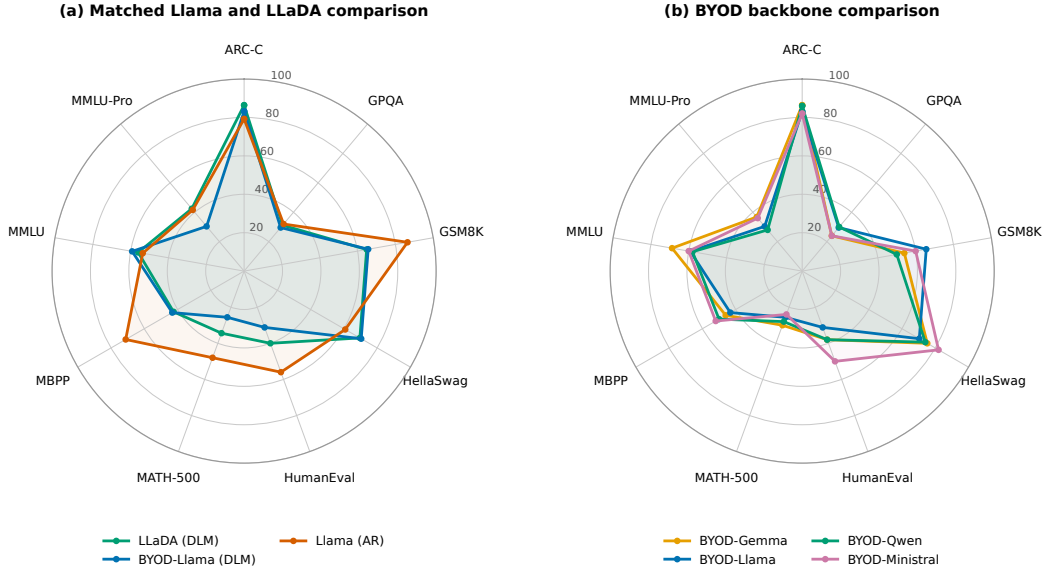


Figure 1: Task-level accuracy profiles from the latest completed evaluations. (a) Matched 125-example evaluations of LLaDA, BYOD-Llama, and the unchanged Llama AR parent. (b) The four primary BYOD conversions; these values summarize the latest completed cells but currently mix 125-example and earlier, generally 50-example, evaluations. Every spoke uses the same absolute 0–100% accuracy scale; benchmark-specific min–max scaling is deliberately avoided because it would make equal radial distances represent different accuracy changes. The polygon areas are descriptive visual summaries, not aggregate scores.

confounder. Every backbone receives the same bidirectional attention intervention, LoRA target modules, and masked-diffusion objective. This paired design separates conversion quality from the strength of the starting checkpoint. We directly compare our models against LLaDA 8B Instruct because its standard masked-diffusion objective, instruction-tuning setup, and overlapping benchmark suite make it the cleanest external control. We additionally report published DiffuLLaMA, Dream-Instruct, iLLaDA-Instruct, and Efficient-DLM values, but keep them protocol-separated: their additional attention, masking, scheduling, data, and inference techniques would confound a controlled comparison.

Our contributions are:

- a single, easily reproducible, architecture-agnostic recipe that converts four instruction-tuned AR model families to masked diffusion using only LoRA parameter updates on a single Google Colab GPU;
- a paired evaluation of each diffusion model and its frozen AR parent across general knowledge, reasoning, mathematics, and code benchmarks;
- a controlled analysis of base-model choice, continued training, denoising-step count, validation loss, and externally scored generation perplexity; and
- a transparent comparison between results produced by our common harness and the published values reported with the LLaDA evaluation protocol.

2 RELATED WORK

Discrete diffusion models corrupt and reconstruct categorical sequences rather than adding Gaussian noise. D3PM formalized categorical corruption with transition matrices, including an absorbing process in which tokens transition to a dedicated mask state (Austin et al., 2021a). MDLM and contemporaneous work showed that this masked formulation admits a simple noise-weighted cross-entropy objective, while RADD reparameterized the predictor in terms of conditional distributions

over clean tokens (Sahoo et al., 2024; Shi et al., 2024; Ou et al., 2025). These approaches form the basis of masked diffusion language models (MDLMs). Uniform (or multinomial) diffusion, which instead corrupts tokens with samples from the ordinary vocabulary, is another option (Hoogeboom et al., 2021); we use the simpler absorbing-mask denoising objective employed by LLaDA.

The authors of LLaDA were the first to show that masked diffusion can scale to an 8B-parameter general-purpose language model (Nie et al., 2025). Although its architecture follows Llama 3.1 8B, LLaDA was pre-trained from scratch using 2.3T tokens and 130,000 GPU-hours. AR-to-diffusion conversion instead amortizes knowledge already acquired by an AR checkpoint. DiffuLLaMA established this route through 65B tokens of full-model continued training at 7B scale (Gong et al., 2025). Dream adopts DiffuLLaMA’s shifted-prediction initialization, continues Qwen2.5 for 580B tokens, and adds context-adaptive token-level noise rescheduling (Ye et al., 2025). Efficient-DLM combines 500B tokens of continued training with blockwise attention and position-dependent masking (Fu et al., 2025), whereas iLLaDA returns to from-scratch diffusion pretraining at substantially larger scale (Nie et al., 2026). Diffusion conversion has also been demonstrated at larger sparse scales by RND1 (30B parameters), LLaDA2.0 (16B and 100B), and DiffusionGemma (25.2B, using uniform diffusion) (Chandrasegaran et al., 2025; Bie et al., 2025; DiffusionGemma Team, 2026).

Hybrid and blockwise models offer a different compromise between bidirectionality and deployability. Block Diffusion is autoregressive across blocks but diffusive within each block, enabling variable-length generation and inter-block KV caching (Arriola et al., 2025). Fast-dLLM v2 converts Qwen2.5 models using full-parameter fine-tuning with approximately 1B tokens and combines blockwise complementary masking with hierarchical caching (Wu et al., 2025). Sequential Diffusion Language Models (SDLM) instead learn next-sequence prediction and dynamically finalize variable-length subsequences, reporting conversion with 2.3 billion tokens (Liu et al., 2025). These systems retain more left-to-right structure than our fully bidirectional answer block, making them especially relevant to speed, caching, and arbitrary-length generation but not suited as direct controls for our intervention.

LAD first showed that LoRA can induce diffusion generation, using a hybrid objective that combines masking with token swaps, duplications, and span shifts (Kuiper et al., 2025; Hu et al., 2022). BYOD instead tests whether a simpler IID absorbing-mask objective can transfer reliably across model families. Other LoRA-based diffusion work addresses different goals: Uno learns diffusion-draft adapters while retaining the AR model as verifier, the dLLM framework uses LoRA to fine-tune already-diffusive checkpoints rather than for its Tiny-A2D conversion, and NaRA adapts existing diffusion models with noise-conditioned LoRA (Sahoo et al., 2026; Zhou et al., 2026; Wang et al., 2026b). Thus, their LoRA components augment already diffusively generating models, rather than converting AR model families into standalone diffusion models.

Table 1 places (estimates of) the training budgets of models of comparable size to our intended conversion targets side by side. For AR-to-diffusion conversion methods, these budgets exclude the substantial earlier cost of training the AR parent; for from-scratch methods they include diffusion pretraining. Dollar and operational-emissions columns are rental-equivalent estimates under one documented pricing and energy scenario, not the authors’ actual bills or measured footprints; Appendix F gives the complete calculation and its limitations.

Our study has explicitly chosen to hold the adaptation mechanism as constant as the architectures permit, repeat it across four families, and benchmark each converted model against the exact AR checkpoint from which it originated. This paired comparison is essential because it allows us to compare our results directly with the prior capability of the parent model.

3 METHOD

3.1 MASKED-DIFFUSION TRAINING OBJECTIVE

The conversion objective teaches the adapted model to recover clean answer tokens from randomly masked versions of the same response while conditioning on an uncorrupted instruction. Equation 1 formalizes this reconstruction loss and its correction for the sampled noise level; the following subsections describe the bidirectional attention intervention and the restricted set of parameters used to optimize it.

Table 1: Incremental training scale and standardized resource estimates for approximately comparable dense 3–9B diffusion systems. “Full” updates the main model weights; LoRA updates diffusion adapters. GPU-hours marked [†] are estimated from published model/data scale; unmarked values are reported. Costs are current cloud-rental equivalents and CO₂e is an operational scenario estimate (Appendix F), not a measured footprint. NR: insufficient information.

System	Diffusion-stage data	Updated weights	Hardware	GPU-h	Cost (USD)	CO ₂ e
BYOD (ours)	≤102.4M tokens	LoRA	1×RTX Pro 6000	3.2	\$11	0.9 kg
DiffuLLaMA 7B	65B tokens	Full	64×GH200	1,917 [†]	\$9,951	0.65 t
LLaDA 8B	2.3T tokens	Full	H800	130,000	\$674,830	44.1 t
Dream 7B	580B tokens + SFT	Full	NR	17,105 [†]	\$88,791	5.81 t
iLLaDA 8B	12T tokens + SFT	Full	NR	404,449 [†]	\$2,099,494	137.3 t
Efficient-DLM 8B	500B tokens	Full	128×H100	16,852 [†]	\$87,479	5.72 t
Uno 8B	7B tokens	LoRA	64×H200	3,840	\$19,933	1.30 t
Fast-dLLM v2 7B	~1B tokens	Full	64×A100	768	\$3,987	149 kg
SDLM 3B	2.3B source tokens	Full	128×GPU (type NR)	NR	NR	NR

Let p be an observed instruction prompt and $x_0 = (x_0^1, \dots, x_0^L)$ its clean response region, where L is the response-region length and x_0^i is its token at position i . For each example, we sample $t \sim \mathcal{U}[10^{-3}, 1]$ and independently replace each eligible response token with a single tokenizer-native mask representation with probability t . Denote the resulting sequence by x_t and the sampled mask positions by M_t . The model is trained with same-position cross-entropy,

$$\mathcal{L}(\theta) = -\mathbb{E}_{t, x_0, x_t} \left[\frac{1}{|E|} \sum_{i \in M_t} \log p_{\phi, \theta}(x_0^i | p, x_t) \right], \quad (1)$$

where E is the set of eligible answer-and-padding positions, $|E|$ is its cardinality, $M_t \subseteq E$ is the subset masked at noise level t , and $p_{\phi, \theta}$ is the model’s conditional token distribution. The expectation is over training examples, sampled noise levels, and their resulting corruptions; ϕ denotes the frozen parent parameters and θ the LoRA parameters. Prompt and chat-template tokens are always visible and never enter the loss. Training corruption is resampled online; validation corruption is deterministic from the example index and seed.

The clean terminating EOS is included as a genuine answer token. To teach the fixed-width generator to emit termination beyond the answer, dynamic batch padding is represented by repeated EOS tokens; these positions are also eligible for masking and loss. Thus padding is part of the learned denoising target rather than conventional ignored padding.

3.2 BIDIRECTIONAL CONVERSION AND TRAINABLE PARAMETERS

We replace the causal mask with an all-zero four-dimensional additive attention mask, so every prompt, answer, and EOS-padding position can attend to every position. The architecture and output vocabulary remain unchanged. This intervention makes right-hand response evidence available to masked positions; Section 5.2 tests whether the converted models use that access in practice. We attach LoRA adapters to every self-attention query and value projection, with rank $r = 1024$, scaling $\alpha = 1024$, and dropout zero. This high rank reflects the scale of the intervention: the adapters must learn a new conditional denoising process rather than a narrow downstream task. In LAD’s rank study, increasing rank from 512 to 1024 at 3B improved every reported benchmark and perplexity, whereas increasing from 1024 to 2048 at 8B gave no consistent further gain; our preliminary rank experiments showed the same pattern. We therefore chose 1024 as a compromise between conversion capacity and training footprint (Kuiper et al., 2025). Backbone weights, token embeddings, the language-model head, biases, and normalization layers are frozen. Each model reuses an existing single-token string from its tokenizer (MASK for Gemma, Llama, and Qwen; <?> for Ministral), avoiding vocabulary expansion.

Table 2 reports the total and trainable parameter counts of the converted models. The trainable fraction varies because the four attention implementations have different projection shapes and layer counts.

After training, we tested whether folding the LoRA layers into the parent projection would change the performance of the model, defined as: $W_{\text{merged}} = W + (\alpha/r)BA$, where W is the frozen base

Table 2: Converted parent checkpoints and audited trainable parameter counts. All trainable parameters are Q/V LoRA weights.

Parent checkpoint	Total parameters	Trainable	Trainable (%)
Gemma 2 9B IT	10.732B	572.5M	5.33
Llama 3.1 8B Instruct	8.466B	436.2M	5.15
Qwen2.5 7B Instruct	7.939B	323.0M	4.07
Ministral 8B Instruct 2410	8.511B	490.7M	5.77

projection, A and B are the trained low-rank adapter matrices, r is their rank, and α is the LoRA scaling parameter. The resulting standalone checkpoint retains bidirectional inference but has the base architecture’s parameter count: Llama decreases from 8.466B adapter-loaded parameters to 8.030B merged parameters.

3.3 TRAINING DATA

All main models use the same training dataset and seed-selected ordering. Examples are retokenized with each parent model’s native chat template and tokenizer. Only the current answer region is corrupted. The dataset builder groups sources into general instruction following, multiple-choice reasoning, mathematics, and code, deduplicates normalized prompts, and blocks exact normalized matches with represented benchmark validation/test prompts.

The four primary models see the same dataset prefix. The continued Llama condition initializes from the primary adapter and uses the next non-overlapping segment of the stable shuffle. It resamples corruption masks and restarts AdamW and its learning-rate schedule; it is therefore continued adapter training, not a restoration of optimizer state. Exact stage sizes are consolidated in Appendix B.

3.4 OPTIMIZATION

We use AdamW with peak learning rate 10^{-5} and train with BF16 master weights plus Transformer Engine HYBRID FP8 execution. All reported conversions ran on a single NVIDIA RTX PRO 6000 Blackwell Server Edition in Google Colab, without multi-GPU or multi-node parallelism; the code, memory strategy, and adapter scope were explicitly designed around this constraint. Appendix B gives the complete optimization, checkpointing, validation, and reproducibility configuration.

3.5 DIFFUSION GENERATION

Generation begins from a fully masked answer block. At each reverse step the model predicts every masked position, after which the least-confident positions are retained for reconsideration according to the LLaDA fixed-transfer schedule; newly accepted positions become permanent context. The primary factual and code benchmarks use greedy token selection and low-confidence remasking. Open-ended evaluation instead samples at temperature 0.7 and delays acceptance of EOS/EOT predictions. Complete task-specific diffusion and AR budgets are reported separately from training details in Appendix C.

4 EVALUATION

4.1 PAIRED MODELS AND BASELINES

For Gemma, Llama, Qwen, and Ministral we compare the diffusion adapter with greedy generation from its unchanged AR parent. LLaDA 8B Instruct is evaluated as an external diffusion baseline. We report two explicitly separated LLaDA comparisons: (i) LLaDA run through our prompts, sampling code, subsets, and scorers; and (ii) values transcribed from the LLaDA paper/evaluation suite. The latter are contextual references rather than directly matched measurements.

4.2 BENCHMARKS AND SCORING

Our benchmark set contains the validation/test sets of MMLU, MMLU-Pro, HellaSwag, ARC-Challenge, GSM8K, MATH-500, GPQA-Main, HumanEval, and sanitized MBPP (Hendrycks et al., 2021a; Wang et al., 2024; Zellers et al., 2019; Clark et al., 2018; Cobbe et al., 2021; Hendrycks et al., 2021b; Rein et al., 2023; Chen et al., 2021; Austin et al., 2021b). We score multiple-choice tasks by option label, GSM8K by its final number, MATH by its normalized final boxed expression, and code tasks by whether the generated program passes the supplied tests in an isolated subprocess.

We evaluate functional use of bidirectional context separately from these capability benchmarks with a deterministic in-place infilling diagnostic (Appendix E). Each example places a fully masked target span between an answer prefix and a suffix that contains the decisive target evidence. All target tokens are masked simultaneously, so the model cannot recover later target tokens from earlier target tokens. We compare the converted model’s one-pass same-position predictions with the suffix present against a prefix-only pass of the same model. For unmerged adapters, we additionally score the untouched causal parent by conventional teacher forcing; this control is deliberately favorable to the AR model but cannot access the right-hand evidence. The diagnostic tests whether the attention intervention is functionally used, rather than measuring broad natural-language reasoning.

For 100 fixed open-ended prompts, Microsoft Phi-4 14B scores each generated answer autoregressively. Our primary aggregate is token-weighted perplexity, $\exp(\sum_j \text{NLL}_j / \sum_j N_j)$, so each scored token contributes equally. Generative perplexity does not penalize loss of sample diversity (Turok et al., 2026). We therefore also compute sliding model-token Distinct-1/2/3 after excluding special tokens. Higher Distinct- n means a larger fraction of unique n -grams.

5 RESULTS

5.1 PRELIMINARY BENCHMARK ACCURACY

Table 3 reports the latest completed matched-harness evaluations. The Llama-25k, Llama-50k, shared Llama-AR, and LLaDA columns use 125 examples per task. The interrupted recovery run reached only Gemma MMLU before stopping, so that cell also uses 125 examples; the remaining Gemma, Qwen, and Ministral cells retain the earlier completed evaluations, generally with 50 examples. These mixed sample counts make the backbone comparison preliminary and unsuitable for significance claims.

The provisional diffusion macro averages are 51.0% for Gemma, 48.7% for Llama-25k, 47.6% for Llama-50k, 49.1% for Qwen, 52.2% for Ministral, and 51.9% for LLaDA in our harness. Because only the Llama and LLaDA columns currently use 125 examples throughout, differences between backbone-wide averages should be treated as descriptive. The matched AR parents outperform their converted counterparts on average, although individual tasks differ. Notably, the second Llama stage does not uniformly improve downstream accuracy despite reducing its 64-step generation perplexity (Section 5.3), motivating checkpoint-wise evaluation rather than selecting solely by denoising loss or fluency. Published values from other diffusion systems are provided separately in Appendix A because their protocols are not matched to ours.

Figure 1 makes the task profiles visible without collapsing them into a macro average. Panel (a) is the cleanest comparison: all three curves use the same nine 125-example subsets and contrast BYOD-Llama with both its unchanged AR parent and LLaDA. The AR parent remains substantially stronger on generation-heavy mathematics and code tasks, whereas the two DLMs have broadly similar profiles. Panel (b) shows that the relative strengths of the four BYOD backbones are task dependent, but remains provisional because the interrupted sweep leaves mixed sample counts across families.

5.2 BIDIRECTIONAL INFILLING DIAGNOSTIC

The separate functional-attention diagnostic directly tests whether target tokens can use decisive evidence to their right (Appendix E). The final paper will summarize its principal result here: **XXX: report the number of converted diffusion models that recover the infill from the suffix, the corresponding prefix-only score, and the causal-parent result.** Detailed per-model exact-match

Table 3: Preliminary exact-match/pass@1 accuracy in our matched harness. D: converted diffusion model; AR: unchanged parent. Llama-50k shares the Llama-25k AR parent. Llama and LLaDA use 125 examples per task; the interrupted remainder of the sweep retains earlier completed values, generally with 50 examples. Among common-harness diffusion columns, bold marks the best result and underline the best converted BYOD result (ties retained); published LLaDA is protocol-separated and excluded from highlighting.

Task	Gemma 25k		Llama			Qwen 25k		Minstral 25k		LLaDA	
	D	AR	D-25k	D-50k	AR	D	AR	D	AR	Ours	Published
<i>General knowledge and reasoning (multiple choice)</i>											
ARC-C	<u>.864</u>	.888	.832	.792	.792	.860	.920	.820	.740	.864	.885
GPQA	<u>.240</u>	.260	.296	.272	.320	<u>.300</u>	.400	.240	.340	.312	.333
HellaSwag	.752	.696	.704	.688	.608	<u>.740</u>	.780	.820	.700	.696	.746
MMLU	<u>.688</u>	.648	.592	.608	.536	.580	.660	.600	.520	.560	.655
MMLU-Pro	<u>.368</u>	.448	.304	.304	.416	.280	.400	.360	.320	.424	.370
<i>Math (open ended)</i>											
GSM8K	.540	.840	<u>.656</u>	.560	.864	.500	.900	.600	.860	.648	.694
MATH-500	<u>.300</u>	.640	.256	.216	.480	.280	.780	.240	.520	.344	.319
<i>Code generation (open ended)</i>											
HumanEval	.380	.740	.312	.336	.560	.380	.960	.500	.940	.400	.494
MBPP	.460	.740	.432	.512	.712	.500	.820	.520	.720	.424	.410
Average	.510	.656	.487	.476	.588	.491	.736	.522	.629	.519	.545

and token-accuracy results are reserved for Appendix Table 9 rather than mixed into the general capability benchmark.

5.3 OPEN-ENDED FLUENCY AND DENOISING BUDGET

Table 4: Open-ended generation at a 128-token output length and 64 NFE for diffusion models. PPL is token-weighted Phi-4 perplexity (lower is better); D-1/2/3 are mean per-answer sliding Distinct- n over model-tokenizer tokens (higher is less repetitive). AR parents generate the same output length autoregressively.

Model family/checkpoint	Diffusion				AR parent			
	PPL↓	D-1↑	D-2↑	D-3↑	PPL↓	D-1↑	D-2↑	D-3↑
Gemma 2 9B, 25k	8.273	.455	.752	.858	3.011	.634	.893	.958
Llama 3.1 8B, 25k	10.503	.527	.802	.905	2.489	.610	.890	.948
Llama 3.1 8B, 50k	8.709	.438	.741	.874	2.489	.610	.890	.948
Qwen2.5 7B, 25k	11.117	.509	.804	.904	2.230	.695	.940	.986
Minstral 8B, 25k	9.424	.493	.786	.888	2.305	.651	.913	.971
LLaDA 8B Instruct	4.065	.514	.823	.916	–	–	–	–

Increasing the denoising budget strongly improves this diagnostic. In the latest open-ended sweep, Llama-25k changes from 19.381 at 32 NFE to 10.503 at 64 and 5.081 at 128; Gemma-25k changes from 14.560 to 8.273 and 4.334. We call settings with fewer denoising network evaluations than generated tokens ($\text{NFE} < L$) the *parallel decoding regime*; unlike AR decoding, they permit more than one token to be finalized per model evaluation. Perplexity values are meaningful only under matched output length, sampler, temperature, and denoising budget.

We also benchmarked the saved Llama adapter before and after merging (Appendix G). Merging removes 436.2M deployed parameters while changing token-weighted perplexity from 19.381 to 19.283 (−0.51%) at 32 NFE and from 3.753 to 3.720 at 128 NFE; Distinct-1/2/3 at 32 NFE change from .495/.788/.899 to .498/.807/.918. On the preliminary nine-task subsets, the unweighted accuracy mean changes from 51.3% to 49.3%, with mixed task-level differences. Across these metric families, merging preserves the model’s overall behavior without a systematic open-generation penalty.

5.4 PROGRESSION WITH TRAINING UPDATES

Figure 2 shows checkpoint-wise generation diagnostics. Improvements are not monotonic at every budget. The matched final-checkpoint comparison suggests that the second Llama stage primarily helps in the parallel decoding regime: for 128-token outputs, 25k to 50k training lowers perplexity from 30.049 to 16.797 at 32 NFE and from 10.503 to 8.709 at 64 NFE, while the 128-NFE value worsens from 3.753 to 4.876. Meanwhile, the preliminary factual-task macro average changes from 48.7% to 47.6%. Continued training therefore appears to improve low-budget parallel decoding rather than general factual capability; ongoing evaluations will test whether this pattern holds across models and seeds. Solid marked lines connect available BYOD checkpoints, and dashed horizontal lines show LLaDA under the same output length and NFE budget. Missing cells are omitted rather than imputed. Panel (b) will test whether changes in perplexity coincide with less repetition rather than treating external-model perplexity as sufficient evidence of quality.

5.5 BASE-MODEL DEPENDENCE AND LLaDA COMPARISON

All four converted families produce usable diffusion generations, but the resulting quality is not architecture invariant. Minstral has the highest provisional BYOD diffusion macro average (52.2%), followed by Gemma (51.0%), while Qwen and Llama are similar on average but differ by task. Figure 1(b) makes these complementary profiles clearer than the macro averages alone. Because the backbone columns currently mix sample sizes, this ordering is descriptive rather than a statistically supported ranking. **XXX: Add confidence intervals and formal paired comparisons once the evaluation size is standardized.**

LLaDA scores 51.9% on the nine-task unweighted average in our 125-example harness, compared with 54.5% from its published values. Agreement varies by task and should not be interpreted as an exact reproduction because prompts, subsets, generation lengths, answer extraction, and code execution may differ. We therefore present our common-harness comparison as primary and Appendix A only as external calibration. We chose LLaDA for direct external testing because it uses

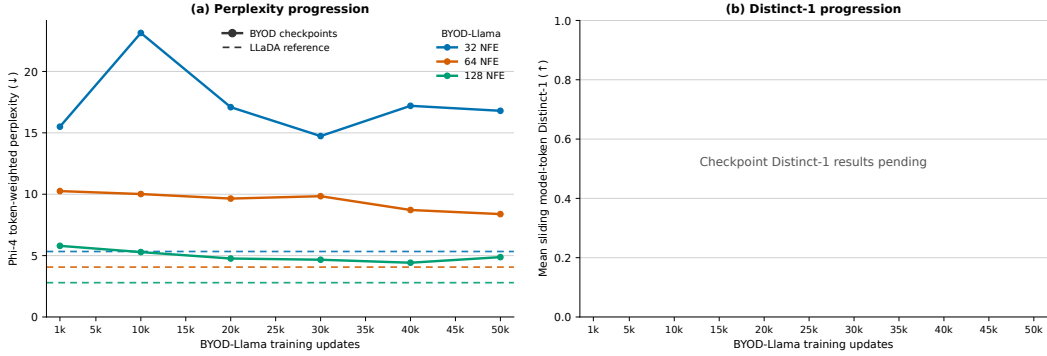


Figure 2: Provisional checkpoint-wise diagnostics for 128-token BYOD-Llama generations, with fixed LLaDA references at 32, 64, and 128 NFE. (a) Token-weighted Phi-4 perplexity (lower is better). (b) Mean sliding model-token Distinct-1 (higher is less repetitive); checkpoint values are pending and the panel is wired to the same workbook-driven plotting pipeline. Solid marked lines denote BYOD checkpoints and color-matched dashed lines denote LLaDA. Blank cells are omitted rather than interpolated.

the uncomplicated response-masking objective closest to ours and reports the largest overlap with our task suite. Dream, iLLaDA, and Efficient-DLM add different data scales, noise weighting, SFT formats, scoring rules, attention patterns, or positional masking. Importing those techniques into the primary experiment would prevent us from attributing outcomes to parameter-efficient conversion itself.

6 DISCUSSION

The central empirical finding is qualitative but robust across the selected runs: high-rank attention-only LoRA is sufficient to turn four independently developed causal instruction models into generators that reconstruct answers from a fully masked block. This supports the view that much of the knowledge needed for diffusion generation is already present in AR representations and vocabulary heads. Because each conversion fits on one Google Colab GPU rather than requiring a dedicated multi-accelerator cluster, the method also lowers the infrastructure threshold for reproducing, modifying, and teaching large-scale diffusion-language-model conversion. A capable paid cloud GPU is still required, so this is broader access rather than zero-cost access.

Our public demos implement EOS-aware early stopping—halting when the pre-EOS answer is identical in two consecutive iterations, together with delayed EOS/EOT retention—which allows actual inference time to fall below the configured fixed-canvas budget; related EOS-led variable-length or early-stopping mechanisms are used by dLLM-Var, VoidPadding, and audio-conditioned diffusion ASR, while JoT instead detects convergence per token (Yang et al., 2025; Liu et al., 2026; Wang et al., 2026a; Kohut et al., 2026).

The merge result also shows that LoRA’s parameter-count overhead is not permanent: the deployed Llama returns from 8.466B to 8.030B parameters without a material open-generation degradation in this small matched comparison (Appendix G). Algebraically folding $(\alpha/r)BA$ into W removes both the stored adapter branch and its extra low-rank inference matrix multiplications. Consequently, using hundreds of millions of high-rank LoRA parameters during conversion need not make the final generator larger or more expensive per network evaluation than the base architecture. This is not model compression—the full base-sized checkpoint remains—and diffusion decoding can still require many network evaluations; more prompts, deterministic parity checks, and backbones are needed to bound small numerical or sampling differences after merging.

Conversion is not free, however. First, 323–573M trainable parameters are parameter-efficient relative to full-model training but substantially larger than conventional low-rank task adapters. Second, our secondary perplexity remains worse than each AR parent and than LLaDA at 64 steps. Third, additional Llama training does not improve the provisional downstream benchmark

aggregate, but its largest fluency gains occur when $NFE < L$. This suggests a more specific benefit: additional denoising training may teach the model to resolve multiple mutually dependent positions per evaluation, improving the quality–compute trade-off without adding corresponding benchmark capability. The 128-NFE result cautions that this interpretation remains preliminary. Finally, EOS-padding supervision makes the fixed-width objective different from a pure answer-only likelihood and deserves explicit ablation.

Perplexity alone is insufficient for selecting a diffusion generator. Our soft-frontier masking ablation lowered Phi-4 generative perplexity but substantially reduced sliding model-token Distinct-1/2/3 and produced visibly more repetitive answers (Appendix H). DiffuLLaMA similarly pairs external-model perplexity with distinct bigrams to check that low perplexity was not obtained through repetition (Gong et al., 2025). More fundamentally, external generative perplexity reflects a separate AR model’s preferences and ignores diversity, so repeated high-probability text can score deceptively well (Turok et al., 2026). Theory likewise distinguishes token-level perplexity from whole-sequence correctness: a masked diffusion model may approach favorable perplexity with few steps while still requiring a step count that scales with sequence length for low sequence error (Feng et al., 2025). We therefore triangulate external perplexity with Distinct-1/2/3, task accuracy, and generated samples. Distinct- n is itself only a repetition diagnostic, so grammaticality, coherence, and actual usability ultimately require systematic qualitative inspection rather than replacement by any single automatic score.

The literature comparison sharpens the trade-off. DiffuLLaMA and Dream reuse AR initialization but update the full model for 65B and 580B tokens, respectively; Efficient-DLM invests 500B tokens and architectural training choices specifically designed to retain causal-model behavior and enable cached block decoding (Gong et al., 2025; Ye et al., 2025; Fu et al., 2025). iLLaDA’s strong reported results come from 12T-token diffusion pretraining plus repeated large-scale SFT, a fundamentally different compute regime (Nie et al., 2026). Fast-dLLM v2 and SDLM show that retaining blockwise or sequential structure can reduce conversion cost while enabling practical caching and variable-length generation (Wu et al., 2025; Liu et al., 2025); our complementary question is how far a fully bidirectional model can be converted with adapter-only updates. Uno is closer to our parameter-efficient philosophy because it also trains LoRA diffusion weights, but its goal is different: an AR verifier guarantees samples from the original AR distribution, whereas BYOD seeks an independently usable fully bidirectional generator (Sahoo et al., 2026). Thus the higher published scores in Appendix A do not isolate a single superior masking objective; they bundle substantially different data budgets and modeling or decoding machinery.

The incremental scale is central to the intended contribution. One primary BYOD conversion processes orders of magnitude fewer token positions than the full-model conversion systems in Table 1, while leaving most parent parameters frozen. This comparison does not erase the sunk cost of AR pretraining, nor prove a proportional reduction in energy or emissions. It does show that an existing foundation checkpoint can be repurposed without repeating that pretraining or reserving a large cluster. That distinction matters both economically and for widening participation in diffusion-model research; direct power and carbon measurements remain necessary for an environmental comparison (Strubell et al., 2019; Patterson et al., 2021).

This is also why we do not claim that Appendix A is a leaderboard. Our direct LLaDA run holds prompts, subsets, extraction, and generation code fixed. The remaining columns are literature context, useful for locating the attainable performance range but unsuitable for causal comparison. A future study can add these systems to the shared harness one at a time, provided their task-specific inference procedures are faithfully reproduced—a requirement emphasized by the dLLM framework (Zhou et al., 2026).

7 LIMITATIONS

The current benchmark sweep is incomplete: Llama and LLaDA use 125 examples per task, whereas the interrupted backbone-wide evaluation retains earlier, generally 50-example, values for many Gemma, Qwen, and Ministral cells. The resulting cross-family estimates have unequal sampling uncertainty and should not support fine-grained ranking. Only one training seed is available per model, so architecture and random-seed effects are not separated. The four backbones differ in tokenizer, pretraining data, alignment, attention implementation, and nominal parameter count. Although chosen

as the latest comparable text-only releases when the study was designed, they are not guaranteed to remain the latest checkpoints at publication time. Phi-4 perplexity favors text that its own AR distribution considers probable, ignores diversity, and is not a diffusion likelihood. Distinct- n detects local repetition but not factuality, grammar, or global coherence. Published LLaDA numbers are not directly comparable to our harness. We also have not yet established wall-clock speedups: $NFE < L$ reduces sequential model calls but does not automatically imply lower latency because each call processes the full answer block. Finally, we did not instrument the runs for energy consumption or carbon intensity. Single-GPU execution and lower incremental training volume support accessibility and compute-efficiency claims, not a quantified sustainability claim.

8 CONCLUSION

BYOD provides a common, parameter-efficient route from instruction-tuned AR checkpoints to masked diffusion generators across Gemma, Llama, Qwen, and Ministral. The preliminary evidence shows broad transfer of the conversion mechanism, but also substantial dependence on the parent model, inference budget, and evaluation metric. By fitting every conversion on a single Google Colab GPU, the work offers a reproducible pathway for researchers who cannot retrain a foundation model or provision a large accelerator cluster. Continued training appears more promising for improving generation in the parallel decoding regime than for improving the aggregate downstream benchmark score, a hypothesis the final checkpoint sweep will test. Completing the paired AR evaluation will determine not merely whether these models can diffuse, but how much of their original capability survives conversion and whether that modest incremental cost is justified relative to purpose-trained diffusion models.

REPRODUCIBILITY STATEMENT

Training and evaluation code is available at <https://github.com/RuurdKuiper/lad-generic>. Public model cards and interactive demos for all four conversions are available under <https://huggingface.co/Ruurd>; each model card links to its corresponding full-precision ZeroGPU demo. The main text specifies the objective, trainable modules, data volume, optimization, generation, benchmarks, and scoring. The accompanying artifacts include resolved configurations, parameter audits, checkpoint identifiers, loss logs, generated samples, benchmark manifests, and frontier-ablation outputs; immutable dataset and software revisions will be recorded with the archival release.

ETHICS STATEMENT

This study uses existing language models and public benchmark or instruction datasets and does not involve human subjects. Converted models inherit factual errors, biases, unsafe behaviors, licensing constraints, and memorization risks from their parent checkpoints and training data. We report aggregate task performance rather than claiming safety or deployment readiness; release artifacts should preserve upstream licenses and document the data provenance and intended research use.

AI USE STATEMENT

Generative AI tools were used to assist software development and debugging, design and review of experiments, creation of analysis scripts, interpretation of preliminary results, literature discovery and summarization, and drafting and editing this manuscript. The authors manually reviewed AI-assisted code and prose, checked numerical claims against saved experimental artifacts, and verified literature claims and citations against primary sources. No benchmark value was entered solely from an unsupported model response. The authors take responsibility for the final content, claims, code, and artifacts.

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A PROTOCOL-SEPARATED LITERATURE RESULTS

Table 5 places our primary Llama conversion beside values reported by other diffusion-model papers (Nie et al., 2025; Ye et al., 2025; Nie et al., 2026; Fu et al., 2025). The BYOD column uses our 125-example common harness, whereas every other column uses its source paper’s evaluation protocol. Differences in prompts, subsets, generation budgets, answer extraction, and execution environments prevent direct ranking; the table is included only to provide scale and task coverage. NR means that the source did not report a value for that task.

Table 5: Our Llama result and protocol-separated published diffusion-model accuracies. Bold marks the largest reported value in each row; it does not imply a controlled comparison. No cross-task averages are shown because task coverage differs by model.

Task	BYOD-Llama (ours)	LLaDA	Dream-Inst.	iLLaDA-Inst.	Efficient-DLM 8B
ARC-C	.832	.885	NR	NR	.619
GPQA	.296	.333	.330	NR	NR
GSM8K	.656	.694	.810	.890	.883
HellaSwag	.704	.746	NR	NR	.725
HumanEval	.312	.494	.555	.659	.683
MATH	.256	.319	.392	.567	.501
MBPP	.432	.410	.588	.580	.610
MMLU	.592	.655	.670	.716	.772
MMLU-Pro	.304	.370	.433	.523	NR

B COMPLETE TRAINING CONFIGURATION

Table 6 records the common resolved configuration of the four primary conversions. The continued Llama condition repeats the same stage from the first adapter checkpoint on the next non-overlapping data segment, with newly sampled masks and a freshly initialized optimizer and schedule; its checkpoints are spaced every 5,000 updates.

Table 6: Full common training configuration. Model-specific parent checkpoints, mask strings, and parameter counts appear in Table 2.

Setting	Value	Setting	Value
Examples per stage	400,000	Optimizer updates per stage	25,000
Maximum sequence length	256	Dynamic padding multiple	16
Batch size	16/16	Gradient accumulation	1
Optimizer	AdamW	Peak learning rate	10^{-5}
Schedule	cosine	Warmup	100 updates
Weight decay	0.01	Maximum gradient norm	0.5
Training precision	FP8 HYBRID	Master/evaluation precision	BF16
Gradient checkpointing	disabled	Selected-logit optimization	enabled
Corruption	IID mask-only	Noise time	$t \sim \mathcal{U}[0.001, 1]$
Mask-eligible region	answer + EOS padding	Loss region	corrupted answer + EOS padding
Answer/padding loss weights	0.9/0.1	Genuine answer EOS	included
LoRA targets	query/value projections	LoRA rank/scaling/dropout	1024/1024/0
Trainable normalization	disabled	Quantization	none
Validation	200 examples/250 updates	Checkpoint interval	1,000 updates
Data-loader workers	4	Prefetch factor	4

C COMPLETE INFERENCE CONFIGURATION

Table 7 separates task-specific diffusion budgets from the training configuration. Unless overridden in the table, diffusion evaluation uses the official LLaDA-style fixed-transfer sampler, greedy predictions (temperature 0), low-confidence remasking, classifier-free guidance scale 0, one full-answer block, and no repetition penalty. Delayed EOS/EOT means that a predicted terminator is assigned the lowest retention confidence and may therefore be predicted but is accepted late; HumanEval instead suppresses terminator logits. The open-ended diagnostic uses temperature 0.7 and 100 fixed prompts. Archived run manifests take precedence over mutable notebook defaults.

Table 7: Task-specific masked-diffusion inference settings in the validation notebook. NFE is the number of denoising network evaluations.

Task	Output tokens	NFE	Block length	Temperature	Terminator handling
Open-ended	128	128/64/32	128	0.7	delayed retention
MMLU	64	64	64	0	ordinary
MMLU-Pro	128	128	128	0	delayed retention
HellaSwag	64	64	64	0	ordinary
ARC-Challenge	64	64	64	0	ordinary
GSM8K	256	256	256	0	delayed retention
MATH-500	256	256	256	0	delayed retention
GPQA-Main	128	128	128	0	delayed retention
HumanEval	256	256	256	0	delayed retention
MBPP	128	128	128	0	delayed retention

Autoregressive parents are decoded greedily. Their maximum-new-token budgets are reported separately because an AR token budget is not an NFE budget and should not be read as computational equivalence.

Table 8: Maximum new tokens for paired greedy AR evaluation.

Task	Tokens	Task	Tokens
Open-ended	128	MMLU	64
MMLU-Pro	2,048	HellaSwag	64
ARC-Challenge	64	GSM8K	256
MATH-500	256	GPQA-Main	128
HumanEval	256	MBPP	128

D INTERACTIVE DENOISING TRAJECTORIES

Figures 3 and 4 illustrate how the public BYOD-Gemma demo exposes the parallel refinement process under random and confidence-guided remasking. All panels use the displayed Amsterdam prompt, the helpful-assistant system prompt, temperature 0.7, top- k 3, initial noise fraction 1.0, delayed EOS/EOT retention, permanent locked retention, and no visualization delay. Early stopping is disabled and block length equals the new-token canvas, ensuring that every panel displays its complete denoising budget. The demo randomizes the hidden sampling seed for every request; we resample the 64-NFE conditions to show readable positive examples, so these trajectories illustrate behavior rather than constitute a controlled quality comparison.

Denoising step 64/64 · 46 output tokens

Amsterdam is the capital of the Netherlands and one of the most popular cities in the world. It is known for its canals, museums, and vibrant nightlife scene. It is also known for its liberal culture and bike-friendly environment.

(a) 64 NFE and 64 tokens.

Denoising step 32/32 · 42 output tokens

Amsterdam is the capital of the country of Netherlands. It is a for-city and is known for the canals, bicycles, and art museums. It is also one of provinces of the Netherlands of the Netherlands.

(b) 32 NFE and 64 tokens.

Denoising step 16/16 · 36 output tokens

Amsterdam is the capital and largest of the Netherlands. It is known for its canals, museums, and and nightlife.. It is a major and center of art and culture..

(c) 16 NFE and 64 tokens.

Figure 3: Confidence-guided BYOD-Gemma-2-9B trajectories at decreasing inference budgets on a 64-token canvas. Red-to-green coloring records each retained token’s sampling probability.

Denoising step 64/64 · 64 output tokens

Amsterdam is the largest and capital city of the Netherlands and is the most populated. It is known for its vibrant culture, rich history, bicycle-friendly culture, and its number of museums and attractions, such as the Rijks, the houses and canals, and the Anne Frank House. It is also a major center for tourism

(a) Random remasking; prediction-iteration coloring.

Denoising step 64/64 · 46 output tokens

Amsterdam is the capital of the Netherlands and one of the most popular cities in the world. It is known for its canals, museums, and vibrant nightlife scene. It is also known for its liberal culture and bike-friendly environment.

(b) Confidence-guided remasking; prediction-iteration coloring.

Figure 4: Remasking comparison for BYOD-Gemma-2-9B at 64 NFE on a 64-token canvas. Blue intensity records the iteration at which each retained token was predicted.

E BIDIRECTIONAL INFILLING DIAGNOSTIC

This diagnostic is designed to isolate access to right-hand evidence, which ordinary left-to-right generation benchmarks do not test directly. It is synthetic and deterministic: no external corpus is downloaded, and no model is fine-tuned on the evaluation examples. The suite contains 100 examples divided evenly between two mechanically generated templates. In the *identifier-copy* subset, a response first leaves an access code blank and a later verification sentence states the code. In the *ordered-relation* subset, a response leaves the label immediately preceding an anchor blank and a later sentence gives the complete local order. Targets are generated alphanumeric identifiers or nonce labels; they are absent from the instruction and answer prefix and occur in the visible suffix. The two templates test the same right-context dependency through different surface forms. They should be interpreted as a controlled functional test, not as a broad benchmark of reasoning, factual knowledge, poetry completion, or free-form infilling quality.

For each model, the instruction is rendered with its native chat template. Let l_k denote the tokenized chat context and answer prefix for example k , let $y_k = (y_k^1, \dots, y_k^{m_k})$ be its target span, and let r_k be the answer suffix. Target length m_k is determined by the model’s own tokenizer. Every target token is replaced simultaneously by the model’s native mask token, producing the full-context input

$$z_k^{\text{full}} = [l_k, \underbrace{\text{MASK}, \dots, \text{MASK}}_{m_k}, r_k]. \quad (2)$$

The converted model performs one bidirectional forward pass and predicts each target from the logits at the same position. There is no iterative sampler, denoising schedule, temperature, top- k filter, or EOS heuristic in this test. Greedy predictions are used only to calculate exact match and token accuracy; target probabilities provide the likelihood diagnostics.

We evaluate two controls. First, the *prefix-only* condition removes r_k while retaining the same prompt, masked target, model weights, and scoring code. The difference between full-context and prefix-only performance therefore measures the benefit obtained from the right-hand evidence within the converted model. Second, when an unmerged PEFT checkpoint is available, we disable its adapter, restore the parent normalization state, reinstate causal attention, and score the clean target with shifted next-token logits. This AR control does not receive the suffix. It is teacher-forced, so target token y_k^j for $j > 1$ may condition on the preceding gold target tokens $y_k^{<j}$; this makes the comparison favorable to the causal parent while preserving its structural inability to condition on r_k . A merged checkpoint cannot recover its untouched parent internally, so its AR control must come from the corresponding unmerged run and is deduplicated when several adapters share that parent.

For condition c , exact match and token accuracy are

$$\text{EM}_c = \frac{1}{K} \sum_{k=1}^K \mathbb{1}[\hat{y}_{k,c} = y_k], \quad (3)$$

$$\text{TokAcc}_c = \frac{\sum_{k=1}^K \sum_{j=1}^{m_k} \mathbb{1}[\hat{y}_{k,c}^j = y_k^j]}{\sum_{k=1}^K m_k}, \quad (4)$$

where $K = 100$, m_k is the target length in model tokens, y_k is the gold target, and $\hat{y}_{k,c}$ is the condition’s position-wise argmax prediction. We also report the token-weighted negative log-likelihood and perplexity,

$$\text{NLL}_c = -\frac{1}{\sum_k m_k} \sum_{k=1}^K \sum_{j=1}^{m_k} \log p_c(y_k^j), \quad (5)$$

$$\text{PPL}_c = \exp(\text{NLL}_c). \quad (6)$$

Suffix gains are $\Delta\text{EM} = \text{EM}_{\text{full}} - \text{EM}_{\text{prefix}}$ and $\Delta\text{TokAcc} = \text{TokAcc}_{\text{full}} - \text{TokAcc}_{\text{prefix}}$; the likelihood gain is $\text{NLL}_{\text{prefix}} - \text{NLL}_{\text{full}}$, so a positive value always denotes useful right context. Aggregate and per-template summaries are retained, together with every tokenized target, prediction, and per-example loss, in the validation artifacts.

Table 9: Bidirectional infilling diagnostic on 100 deterministic examples. Full and prefix-only columns evaluate the converted model; Δ suffix is full-context minus prefix-only exact match. AR is the untouched causal parent under favorable teacher-forced scoring. EM: exact match; Tok.: token accuracy. All values are pending.

Model	Full EM	Full Tok.	Prefix EM	Δ suffix EM	AR EM
Gemma 2 9B, 25k	XXX	XXX	XXX	XXX	XXX
Llama 3.1 8B, 25k	XXX	XXX	XXX	XXX	XXX
Llama 3.1 8B, 50k	XXX	XXX	XXX	XXX	XXX
Qwen2.5 7B, 25k	XXX	XXX	XXX	XXX	XXX
Ministral 8B, 25k	XXX	XXX	XXX	XXX	XXX

F COST AND OPERATIONAL-EMISSIONS ESTIMATION

Table 1 compares the cost of training a model that produces diffusion behavior. It should thus be taken into account that some of these models build on an AR parent, whereas others are trained from scratch. It also excludes exploratory runs, failed jobs, dataset construction, evaluation, storage, and inference. The estimates must therefore not be interpreted as total project or lifecycle footprints.

For BYOD we use the observed 3.2 GPU-hours for the conversion of a Llama 3.1 8B model for diffusive generation. For a prior work that reports GPU-hours, we retain that value. Where a paper reports model size but not time, we approximate training compute as $C \approx 6ND$ FLOPs (Hoffmann et al., 2022), where C is total training compute, N is the number of model parameters, and D is the number of processed training-token positions. We then calculate

$$G_{\text{est}} = \frac{6ND}{0.40 (989 \times 10^{12}) 3600}, \quad (7)$$

where G_{est} is aggregate estimated GPU-hours, 0.40 is the assumed model FLOPs utilization (MFU), 989×10^{12} is the NVIDIA H100’s dense BF16 Tensor Core throughput in FLOPs/s (NVIDIA, 2023), and 3600 converts seconds to hours. The 40% MFU assumption is consistent with the 39.0–45.0% BF16 MFUs reported by FP8-LM for GPT-style 7B, 13B, and 175B training on H100 GPUs (Peng et al., 2023). Actual utilization varies considerably with architecture, sequence length, parallelization, communication, activation recomputation, and implementation efficiency. The assumption is therefore a standardized order-of-magnitude convention rather than a reconstruction of any authors’ cluster. It yields the daggered values in Table 1. The Dream and iLLaDA estimates cover their main 580B- and 12T-token stages but omit the smaller instruction-tuning additions, so they are lower bounds.

Table 10 calibrates the same calculation against studies that report both a usable token count and elapsed accelerator-hours (Nie et al., 2025; Wu et al., 2025). For LLaDA, it predicts approximately 77,519 GPU-hours, compared with 130,000 reported H800 GPU-hours, corresponding to an implied utilization of 23.9%. Thus 40% can be optimistic about achieved utilization and correspondingly conservative with respect to estimated accelerator-hours, rental cost, and emissions. Fast-dLLM v2 displays a still larger gap. Its specialized training concatenates clean and noised views and uses a nonstandard attention pattern, illustrating that an “implied MFU” from $6ND$ is a calibration diagnostic, not a measured hardware-utilization statistic.

Table 10: Calibration of the $6ND$ estimator against reported accelerator-hours. Predictions use 40% MFU and the stated accelerator’s dense-BF16 peak (989 TFLOP/s H100-class proxy for H800; 312 TFLOP/s for A100) (NVIDIA, 2023; 2021). “Implied MFU” is the utilization that would reconcile $6ND$ with reported time; it is not a measured MFU.

System	Parameters	Tokens	Pred. GPU-h	Reported GPU-h	Implied MFU
LLaDA	8B	2.30T	77,519	130,000	23.9%
Fast-dLLM v2	7B	1.31B	123	768	6.4%

We do not apply this estimator to SDLM. Its released 3B training script launches 128 GPU workers, but neither accelerator type nor elapsed training time is reported (Liu et al., 2025). More importantly,

SDLM’s 2.3B figure describes source-corpus tokens: its parallel-block construction rearranges each example and may append multiple noise/target blocks within a 5,632-token training sequence. Consequently, substituting 2.3B directly for D would undercount the token positions actually processed, while the $6ND$ approximation would also omit the material sequence-length-dependent attention cost. We therefore report its known GPU count but leave GPU-hours, cost, and emissions as NR rather than present a misleading lower-bound estimate.

We convert BYOD time to a rental-equivalent cost using the September 2026 AWS US-East on-demand price of \$3.36 per hour for a one-GPU `g7e.2xlarge`, which contains the same 96-GB NVIDIA RTX PRO Server 6000 accelerator used in our Colab run (Amazon Web Services, 2026b). For other systems we use one consistent current major-cloud proxy: \$5.191 per accelerator-hour for an AWS `p5.4xlarge` H100 Capacity Block in US East (Amazon Web Services, 2026a). Thus these figures answer “what would the stated or estimated accelerator-hours cost at a public contemporary rate?”, not what the original authors paid. They exclude storage, egress, idle reservation time, taxes, and discounts.

Operational emissions are estimated as energy use multiplied by the data center’s power usage effectiveness (PUE) and the grid carbon intensity (Lannelongue et al., 2021; Patterson et al., 2021):

$$\hat{E}_{\text{CO}_2\text{e}} = G P_{\text{GPU}} \text{PUE} I_{\text{grid}}. \quad (8)$$

Here $\hat{E}_{\text{CO}_2\text{e}}$ is estimated operational emissions in kg CO_2e , G is aggregate GPU-hours, P_{GPU} is assumed per-GPU power draw in kW, PUE is the dimensionless data-center power usage effectiveness, and I_{grid} is grid carbon intensity in kg $\text{CO}_2\text{e}/\text{kWh}$. We use 0.60 kW for RTX PRO 6000 runs, matching NVIDIA’s stated maximum board power (NVIDIA, 2025), and a 0.70-kW H100-class proxy for all other entries except the reported A100 run, for which we use 0.40 kW (NVIDIA, 2023; 2021). We assume PUE 1.09, Google’s reported fleet-wide value (Google Cloud, 2025), and the IEA’s 2024 global-average generation intensity of 0.445 kg $\text{CO}_2\text{e}/\text{kWh}$ (International Energy Agency, 2025). For example, one BYOD stage gives $3.2 \times 0.60 \times 1.09 \times 0.445 = 0.93$ kg CO_2e . These are location-independent GPU-only operating scenarios: they assume full rated GPU power, omit host CPU/memory/network energy and embodied hardware emissions, and do not account for renewable-energy procurement. Direct power telemetry and the datacenter’s time- and location-specific marginal carbon intensity would be required for a measured footprint.

G ADAPTER-MERGING ABLATION

Table 11: Open-generation comparison of the adapter-loaded and merged Llama checkpoints. PPL is token-weighted; Distinct- n is averaged over answers. Both use 128 generated tokens; D-1/2/3 use 32 NFE.

Representation	Parameters	PPL ₃₂ ↓	PPL ₁₂₈ ↓	D-1↑	D-2↑	D-3↑
Adapter loaded	8.466B	19.381	3.753	.495	.788	.899
Merged standalone	8.030B	19.283	3.720	.498	.807	.918

Table 12: Factual benchmark accuracy (%) before and after merging. These are preliminary 50-example task subsets; the mean is unweighted across the nine tasks.

Task	Adapter	Merged	Task	Adapter	Merged
ARC-C	82.8	82.0	MATH	22.0	22.0
GPQA	28.0	30.0	MBPP	46.0	44.0
GSM8K	65.2	58.0	MMLU	61.2	64.0
HellaSwag	73.2	68.0	MMLU-Pro	33.2	28.0
HumanEval	50.0	48.0	Mean	51.3	49.3

Merging is an algebraic reparameterization rather than an additional training stage: each base projection W is replaced by $W + (\alpha/r)BA$, where A and B are the trained rank- r LoRA matrices and α is their scaling parameter. Subject to floating-point rounding, evaluating the folded projection is equivalent to evaluating the base and LoRA branches separately. The open-generation metrics are

slightly better after merging, whereas the small factual subsets show mixed task differences and a 2.0-point lower macro mean. These reruns therefore support preservation of overall behavior, but are not precise enough to attribute task-level changes to merging; a logit-level deterministic parity test would isolate numerical merge error from generation and subset variance.

H FRONTIER-MASKING ABLATION

The selected main runs use the simpler IID corruption process: after sampling the example-level noise fraction t , each eligible answer or EOS-padding position is independently masked with probability t . In a later ablation, we tested *soft-frontier masking*, intended to resemble partially completed left-to-right text while retaining stochastic exceptions. For a frontier example, masking probability increases sigmoidally with answer position around a frontier determined by t ; the tested configuration used a 0.75 probability of selecting this branch per example, $\epsilon = 0.03$, and $\tau = 3.0$, with the remaining examples using IID corruption. Token losses use the corresponding inverse masking-probability correction.

The frontier variant achieved lower external Phi-4 generative perplexity than the matched IID variant, but substantially lower sliding model-token Distinct-1, Distinct-2, and Distinct-3, accompanied by visibly more immediate token and phrase repetition. This disagreement is precisely why the main experiments retain IID corruption and why generative perplexity is not used as the sole selection criterion. **XXX: Insert the matched run identifiers, checkpoint, generation NFE, token-weighted perplexity, Distinct-1/2/3 values, and representative blinded samples after consolidating the ablation artifacts.**